CROSS-MODAL KNOWLEDGE DISTILLATION FOR LARGE-SCALE EEG CLASSIFICATION ON THE THINGS-EEG DATASET

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ABSTRACT

Decoding visual semantics from EEG is challenging due to noisy signals and weak class separability, particularly under large label spaces. We study a 1654-way supervised classification task on the *seen* split of THINGS-EEG-Text and propose a cross-modal knowledge distillation paradigm. A multi-modal teacher trained on image and text features provides softened targets to guide an EEG-only student during training, while inference remains EEG-only. Experiments show that distillation consistently improves accuracy over EEG-only baselines, achieving more than a two-fold relative gain in Top-1 test accuracy under the best hyperparameters.

Keywords EEG decoding · Knowledge Distillation · Multi-modal Learning · Large-scale Classification

1 Introduction

Traditional machine learning (ML) methods are typically developed under the assumption that training and test data are drawn from the same distribution and that all target classes are fully observed during training. Under this paradigm, models rely heavily on fixed feature representations and supervised learning with hard labels, which limits their ability to generalize beyond seen data. While such approaches have achieved success in well-controlled settings, they often struggle in scenarios involving high-dimensional inputs, weak or noisy signals, or limited supervision.

In contrast, advanced machine learning paradigms aim to overcome these limitations by enabling models to transfer knowledge, reason beyond observed categories, and adapt to more complex environments. Zero-shot learning (ZSL) addresses this limitation by leveraging semantic information to recognise unseen classes. Long et al. [2017]demonstrate that synthesising visual features enables generalisation beyond the training label space.

Recent advances in deep learning have enabled effective knowledge transfer across tasks and modalities, providing the representational capacity required for cross-modal learning.

Inspired by these advances, this project adopts a knowledge transfer perspective to address the challenge of EEG-based classification. Electroencephalography (EEG) signals are inherently noisy, exhibit low signal-to-noise ratios, and provide weak class-discriminative structure compared to modalities such as images or text. Directly training supervised models on EEG data alone often leads to limited performance, particularly in large-scale classification settings. To mitigate this issue, we employ a knowledge distillation framework, where a strong multi-modal teacher model—trained on image and text representations—is used to guide a weaker EEG-based student model. Knowledge distillation, originally introduced by Hinton et al. [2015], transfers soft target distributions from a teacher to a student, enabling the student to capture inter-class similarity structure that is not available from hard labels alone.

By distilling knowledge across modalities, our approach bridges traditional supervised learning and advanced ML paradigms that emphasise knowledge transfer and representation learning. Rather than explicitly performing zero-shot learning or rule generation, this work demonstrates how principles from advanced machine learning—particularly semantic knowledge transfer—can be practically applied to improve performance in a challenging real-world setting. The proposed framework provides a principled and efficient way to leverage strong auxiliary modalities to enhance EEG classification, highlighting the broader potential of advanced ML techniques in multi-modal and weak-signal domains.

2 Data and Paradigm

2.1 Task Setting

In this work, we focus on supervised EEG-based object classification using the *seen* split of the ThingsEEG-Text dataset. The seen split contains 16,540 EEG trials spanning 1,654 object categories, forming a large-scale fixed-label multi-class classification problem (Figure 1a). Since the objective of this work is to analyse knowledge transfer under a controlled supervised paradigm, all evaluations are restricted to the seen split.).

2.2 Data Exploration and Splitting Strategy

EEG signals are inherently noisy and weakly discriminative at the single-trial level. As shown in Figure 2, EEG feature distributions exhibit large variance and substantial overlap across samples, indicating limited class separability.

PCA projections further reveal heavy inter-class overlap in EEG representations(Figure 3), whereas semantic embeddings derived from image and text modalities exhibit clearer clustering structure (Figures 4a and 4b).

Representational similarity analysis using cosine-based RDMs shows that EEG representations encode weaker semantic relationships compared to image and text features(Figures 5a–5c).

To ensure reliable and unbiased evaluation, we adopt a stratified split of the seen data into training, validation, and test sets with a ratio of 60/20/20. All splits share the same label space, and class proportions are preserved across splits.

The training set is used to learn model parameters, the validation set is employed for hyperparameter tuning and early stopping, and the test set is reserved exclusively for final performance reporting. This strategy ensures fair comparison across models while preventing information leakage.

2.3 Learning Paradigm and Baseline Definition

Given the limited discriminative power of EEG features, directly training models on EEG alone leads to poor generalisation. We therefore adopt a simple EEG-only linear classifier as a baseline to establish task difficulty.

In contrast, the strong semantic structure in image and text embeddings motivates leveraging auxiliary modalities during training, while retaining EEG-only inference.

2.4 Knowledge Distillation Paradigm and Workflow

Based on the above observations, we adopt a cross-modal knowledge distillation paradigm. A teacher model trained on image and text embeddings provides semantic supervision to an EEG-based student model, which is trained to perform the same classification task using EEG inputs only.

During training, the student is supervised by both ground-truth labels and soft targets from the teacher, enabling the transfer of inter-class semantic relationships into EEG representations. At inference time, only the EEG-based student model is used.

Figure 6 illustrates the training–inference workflow, highlighting that multi-modal information is exploited only during training, while deployment relies solely on EEG signals.

2.5 Paradigm Formalisation and Real-World Value

At a high level, the student training objective combines hard-label supervision with knowledge distillation:

$$\mathcal{L} = (1-\alpha)\mathcal{L}_{CE} + \alpha\mathcal{L}_{KD}, \tag{1}$$

where $\mathcal{L}_{CE}$ denotes the cross-entropy loss with ground-truth labels and $\mathcal{L}_{KD}$ encourages alignment between the student and teacher output distributions.

Under this learning paradigm, EEG-only inference is strictly enforced, while semantic-rich image and text modalities are exploited solely during training. Data exploration reveals severe class overlap and weak semantic structure in EEG representations, making hard-label supervision insufficient for large-scale classification. Cross-modal knowledge distillation therefore provides a principled mechanism to transfer inter-class semantic relationships without increasing deployment cost.

The proposed paradigm provides clear practical benefits for real-world EEG-based recognition systems. Auxiliary modalities are restricted to the training phase, ensuring low deployment cost and making the approach suitable for

realistic brain–computer interface scenarios. By transferring semantic knowledge from visual–textual representations, the EEG-based student model becomes more robust to noise and limited supervision, leading to improved ranking performance and better-calibrated predictions.

3 Model Development

3.1 Problem Formulation

We consider a supervised multi-class classification problem on the seen EEG dataset. Each EEG sample is represented as a feature vector $x \in \mathbb{R}^{561}$, with an associated class label $y \in \{1, \ldots, C\}$, where $C = 1654$ denotes the number of seen object categories.

The objective is to learn a parametric classifier $f_\theta : \mathbb{R}^{561} \to \Delta^C$, which maps EEG features to a categorical probability distribution over C classes.

3.2 Baseline Model (A): EEG-only Supervised Learning

As a reference model, we adopt an EEG-only supervised learning baseline. Let $p_\theta(y \mid x)$ denote the predicted class probability. The baseline model is trained by minimizing the standard cross-entropy loss:

$$\mathcal{L}_{CE}(\theta) = -\mathbb{E}_{(x,y)} \left[\log p_\theta(y \mid x) \right]. \tag{2}$$

EEG features are standardized prior to classification, and optional dimensionality reduction via Principal Component Analysis (PCA) is applied to suppress noise and redundancy. This baseline relies solely on hard-label supervision and establishes a lower-bound reference for EEG-based classification performance.

3.3 Methodological Improvement (B): Knowledge Distillation

3.3.1 Motivation

EEG signals are noisy and exhibit weak class separability, particularly in large-scale classification settings with limited samples per class. Hard-label supervision provides only a single target per sample and does not encode inter-class similarity.

In contrast, models trained on image and text features capture richer semantic relationships between object categories. Knowledge distillation provides a principled mechanism to transfer such semantic information from a stronger teacher model to a weaker EEG-based student model during training.

3.3.2 Distillation Objective

Let z_s and z_t denote the logits produced by the student and teacher models, respectively. Softened output distributions are obtained using a temperature parameter $T > 1$:

$$p_s^T = \mathrm{softmax}\left(\frac{z_s}{T}\right), \quad p_t^T = \mathrm{softmax}\left(\frac{z_t}{T}\right). \tag{3}$$

The distillation loss is defined as the Kullback–Leibler divergence between the teacher and student distributions:

$$\mathcal{L}_{KD} = T^2 \cdot \mathrm{KL}\left(p_t^T \,\|\, p_s^T\right). \tag{4}$$

The student model is trained using a weighted combination of cross-entropy and distillation losses:

$$\mathcal{L}_{\text{total}} = (1-\alpha)\mathcal{L}_{CE} + \alpha \mathcal{L}_{KD}, \tag{5}$$

where $\alpha \in [0, 1]$ controls the contribution of distilled knowledge. Setting $\alpha = 0$ reduces the objective to the EEG-only baseline.

3.3.3 Theoretical Expectations

By introducing soft target distributions, knowledge distillation provides richer supervisory signals that encode inter-class similarity. The KL divergence term acts as an implicit regularizer, discouraging over-confident predictions and improving generalization in noisy and high-dimensional EEG feature spaces.

The teacher model is used only during training, while inference relies exclusively on EEG inputs.

3.4 Proposed Model (A+B)

The proposed model combines the EEG-only baseline classifier (A) with the distillation-based training strategy (B). The student model takes EEG features as input, while a teacher model trained on concatenated image and text embeddings provides soft supervisory signals during training. Knowledge transfer is performed at the output distribution level, requiring no alignment between internal representations.

3.5 Training Strategy

Student model parameters are optimized by minimizing $\mathcal{L}_{\text{total}}$ using mini-batch stochastic optimization. Hyperparameters, including the temperature T and weighting coefficient α, are selected based on validation performance. Early stopping is applied to mitigate overfitting.

At test time, only the EEG-based student model is used, ensuring that inference does not require auxiliary modalities.

4 Result Analysis

This section evaluates whether the proposed knowledge distillation (KD) paradigm effectively addresses the challenges identified in earlier analysis, namely weak class separability and limited semantic structure in EEG representations. Quantitative ablation studies, training dynamics, and qualitative analyses are conducted under a unified evaluation protocol.

4.1 Quantitative Evaluation and Ablation Study

We evaluate multiple models on the seen EEG classification task, including an EEG-only baseline, an EEG student trained with hard labels only, and EEG students trained with knowledge distillation from a multi-modal teacher. The teacher model trained on image and text features is reported as an upper-bound reference.

Table 2 summarizes the Top-1 test accuracies under different KD configurations. The EEG-only baseline achieves extremely low accuracy, reflecting the intrinsic difficulty of large-scale EEG classification. Training an EEG student with cross-entropy loss yields only limited improvement, indicating that increased model capacity alone is insufficient.

In contrast, incorporating knowledge distillation consistently improves performance across all evaluated settings. The best configuration ($T = 8.0$, $\alpha = 0.7$) achieves a validation accuracy of 4.26% and a test accuracy of 3.45%, more than doubling performance compared with the EEG-only student. The teacher model achieves substantially higher accuracy, confirming that image and text modalities encode rich semantic structure. These results directly validate the theoretical expectation that soft target distributions provide richer supervision than hard labels alone.

4.2 Training Dynamics and Hyperparameter Sensitivity

Figure 7 shows validation accuracy over training epochs. Compared to the EEG-only student, the KD-trained model converges faster and consistently achieves higher validation accuracy, supporting the hypothesis that soft targets facilitate more effective learning.

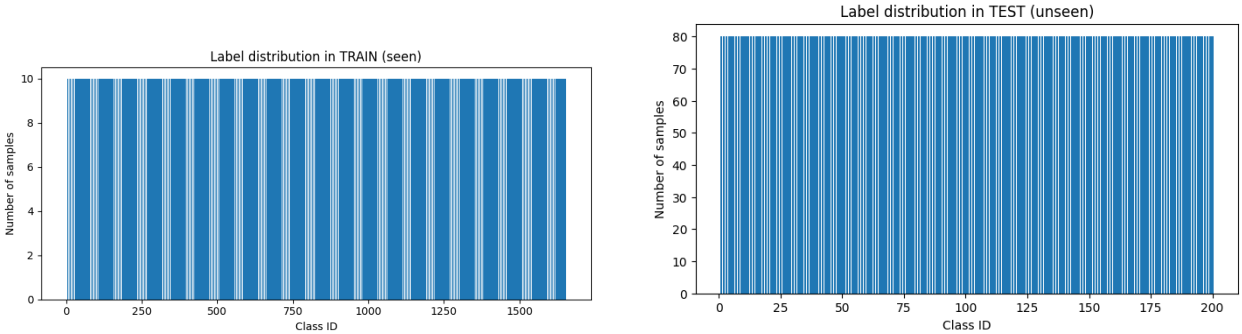
Figure 8 presents the corresponding training loss curves. The EEG-only model exhibits rapid loss reduction, suggesting potential overfitting in a high-dimensional label space. In contrast, the KD model shows smoother and more stable optimization behavior, consistent with the role of knowledge distillation as an implicit regularizer.

4.3 Qualitative Analysis and Failure Diagnosis

Figure 9 compares Top-k test accuracy between the EEG-only baseline and the best KD model. While absolute Top-1 accuracy remains low due to the large number of classes ($C = 1654$), the KD model achieves notable gains in Top-5 and Top-10 accuracy, indicating improved semantic ranking.

Figure 10 shows the distribution of maximum softmax confidence scores. The baseline model tends to produce overconfident predictions despite low accuracy, whereas the KD model exhibits a more conservative and better-calibrated confidence distribution.

Due to the large label space, a full confusion matrix is not interpretable. Figure 11 therefore presents row-normalized confusion matrices on a representative subset of 50 classes. Compared to the baseline, the KD-trained model exhibits a clearer diagonal structure and reduced systematic confusions, indicating improved semantic alignment.



(a) Label distribution of the seen split

(b) Label distribution of the unseen split

Figure 1: Class distributions of the seen and unseen splits in the ThingsEEG-Text dataset. The seen split contains 1,654 object categories with balanced class frequencies, forming a large-scale fixed-label classification task used in this work. Although the unseen split (b) follows a different label space, standard supervised metrics are not directly applicable under a fixed-classifier setting; therefore, all experiments are conducted on the seen data only.

Table 1: Stratified train/validation/test split of the seen dataset. Class proportions are preserved across splits to ensure unbiased evaluation and fair model comparison.

Split	Proportion	Purpose
Train	60%	Model training
Validation	20%	Hyperparameter tuning
Test	20%	Final evaluation

Table 2: Ablation study of knowledge distillation (KD) on the seen EEG classification task.Validation accuracy is used for hyperparameter selection; test accuracy is reported for final comparison. All numbers are Top-1 accuracy (%).

Model	T	α	Val (%)	Test (%)
EEG Linear Baseline (Std + PCA + LogReg)	–	–	1.48	1.54
EEG Student (CE only)	–	0.0	1.90	1.69
EEG Student + KD	2.0	0.3	2.24	–
EEG Student + KD	2.0	0.5	2.45	–
EEG Student + KD	2.0	0.7	2.93	–
EEG Student + KD	4.0	0.3	2.00	–
EEG Student + KD	4.0	0.5	3.57	–
EEG Student + KD	4.0	0.7	3.69	–
EEG Student + KD	8.0	0.3	2.51	–
EEG Student + KD	8.0	0.5	3.23	–
EEG Student + KD (best)	**8.0**	**0.7**	**4.26**	**3.45**
Teacher (Image + Text)	–	–	67.26	66.41

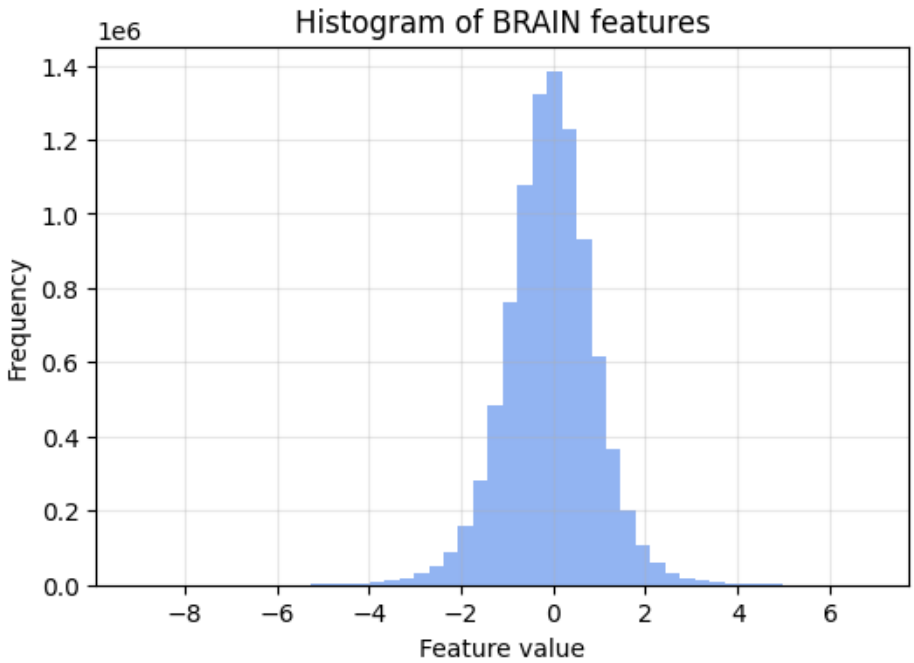


Figure 2: Histogram of EEG feature values, showing wide variance and heavy overlap across samples. The wide variance and heavy overlap indicate weak discriminative structure and low signal-to-noise ratio in EEG representations.

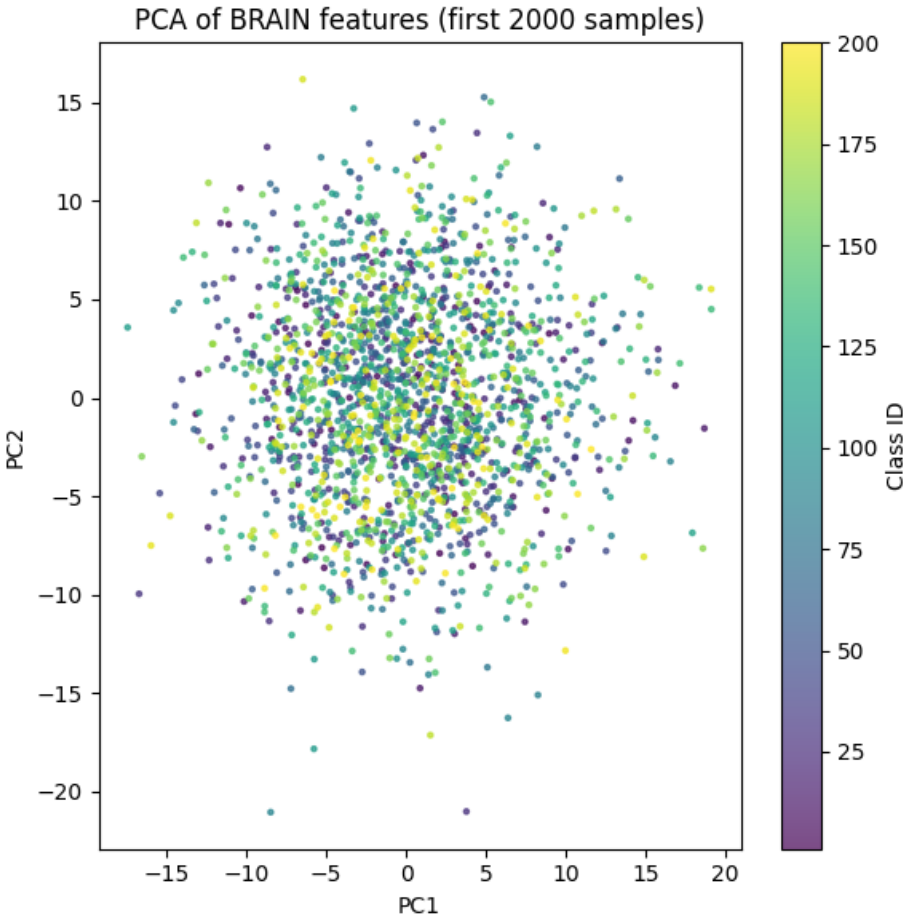
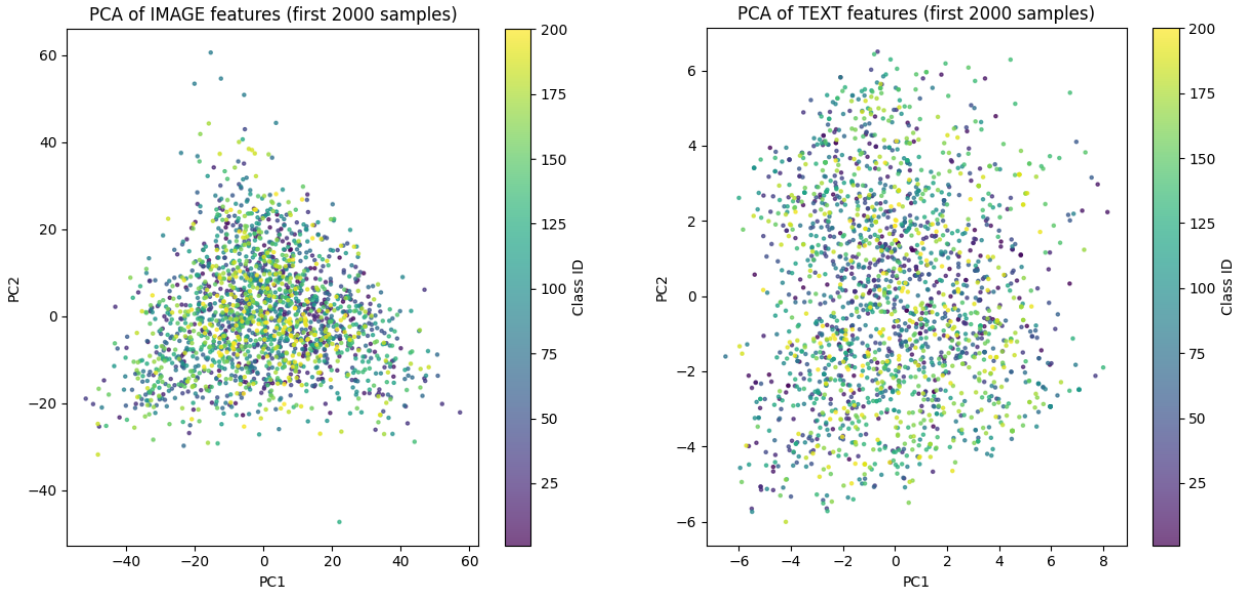


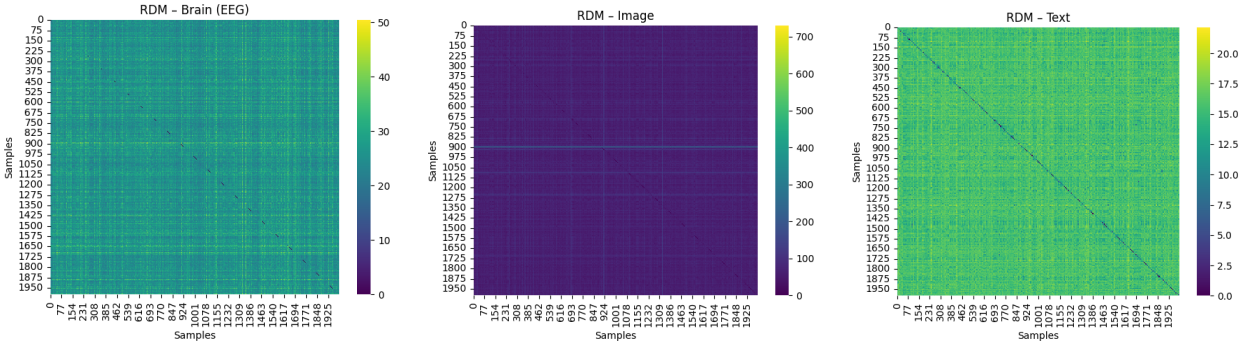
Figure 3: PCA projection of EEG features. Heavy inter-class overlap suggests poor class separability in low-dimensional EEG representations.



(a) PCA of image embeddings

(b) PCA of text embeddings

Figure 4: Low-dimensional projections of semantic-rich image and text representations. Compared to EEG features, semantic-rich modalities exhibit clearer clustering structure, indicating stronger inter-class semantic organization.



(a) EEG RDM

(b) Image RDM

(c) Text RDM

Figure 5: Representational dissimilarity matrices (RDMs) for (a) EEG, (b) image, and (c) text representations. EEG exhibits weaker block-diagonal structure, indicating limited encoding of semantic relationships compared to image and text modalities.

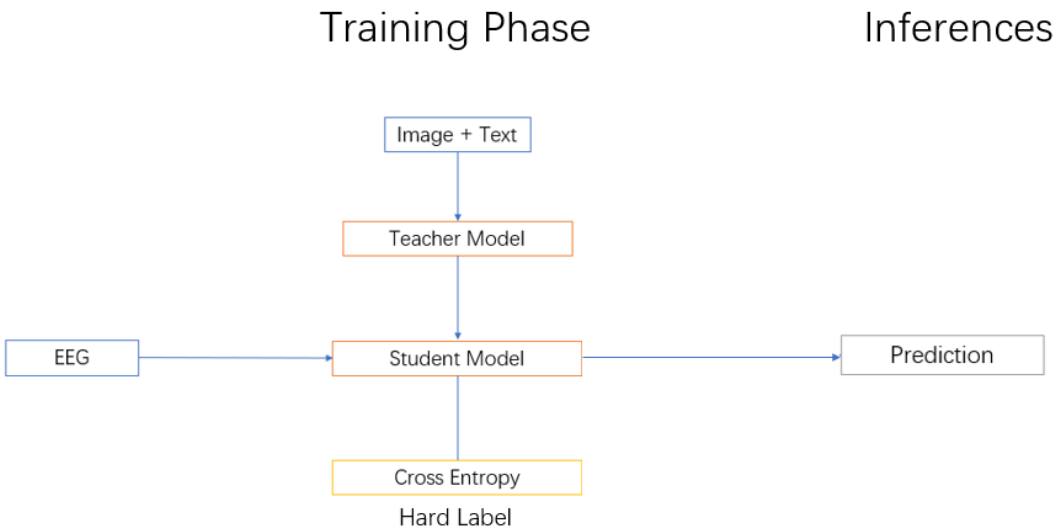


Figure 6: Cross-modal knowledge distillation workflow. Image and text embeddings are used only during training to supervise an EEG-based student via soft targets, while inference strictly relies on EEG signals, ensuring practical deployability and no additional sensing cost.

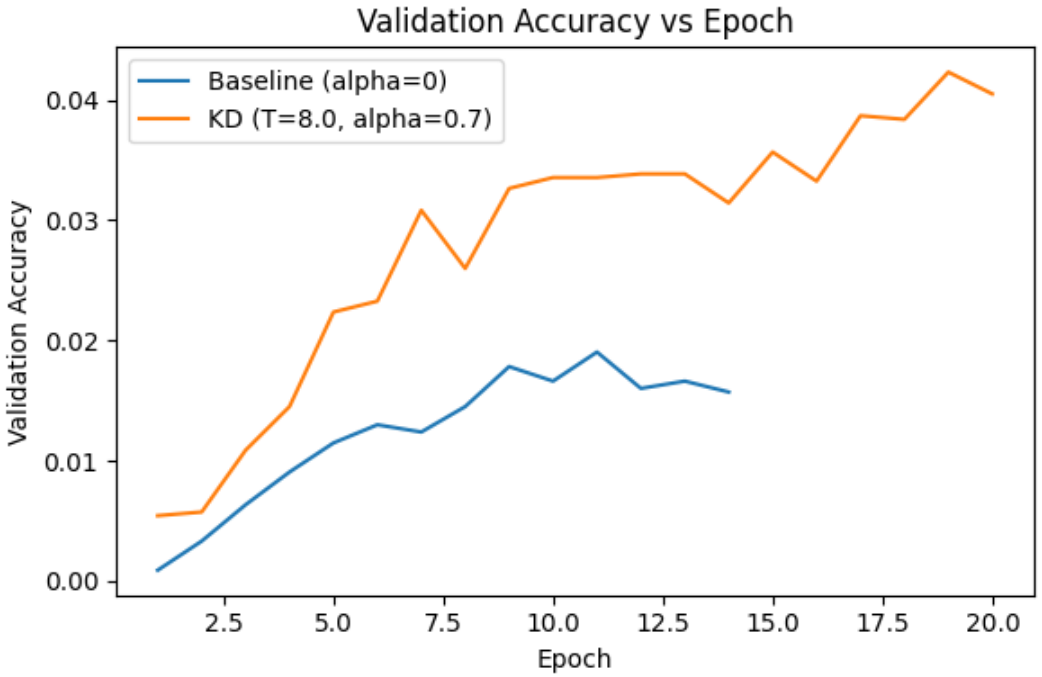


Figure 7: Validation accuracy versus training epochs for the EEG-only student and the knowledge-distilled student. Knowledge distillation leads to faster convergence and consistently higher validation accuracy.

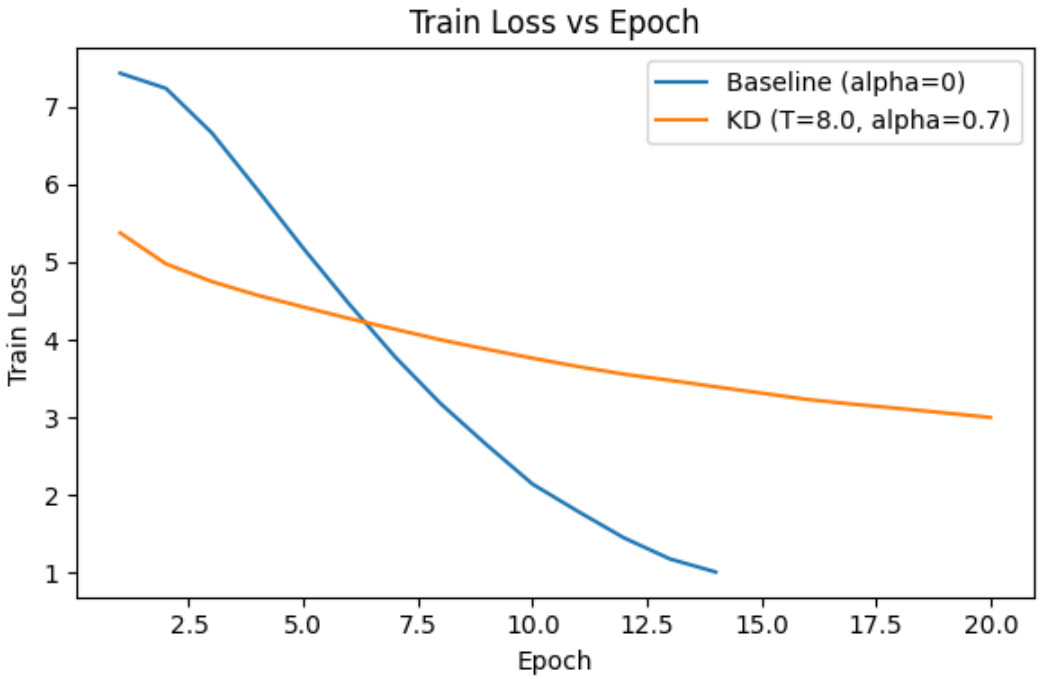


Figure 8: Training loss curves for the EEG-only student and the knowledge-distilled student.The smoother optimization trajectory of the KD model is consistent with the regularization effect of soft targets.

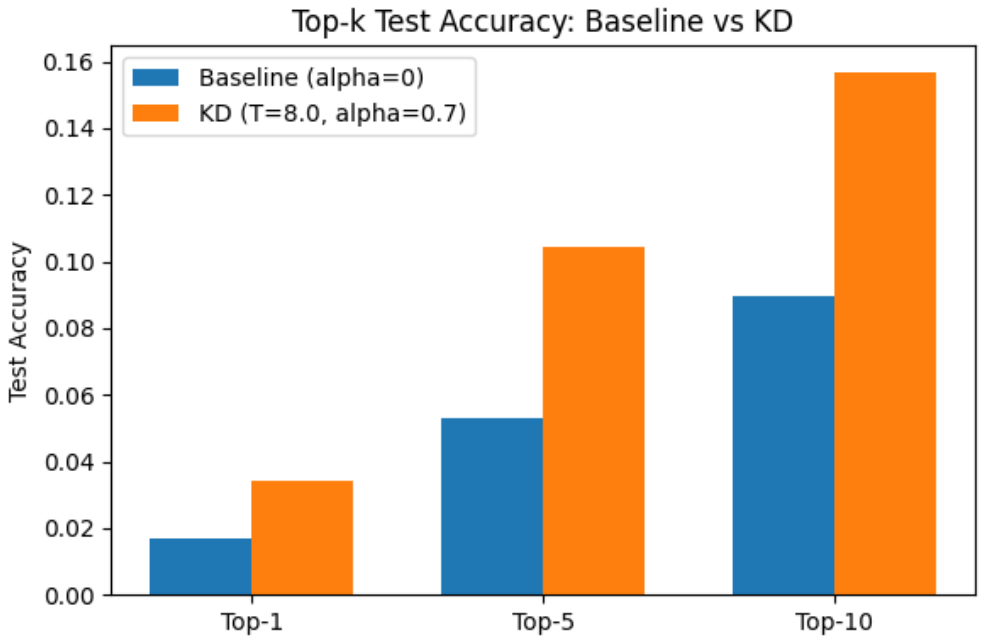


Figure 9: Top-k test accuracy comparison between the EEG-only baseline and the best-performing knowledge-distilled model. Knowledge distillation yields larger gains in Top-5 and Top-10 accuracy, indicating improved semantic ranking.

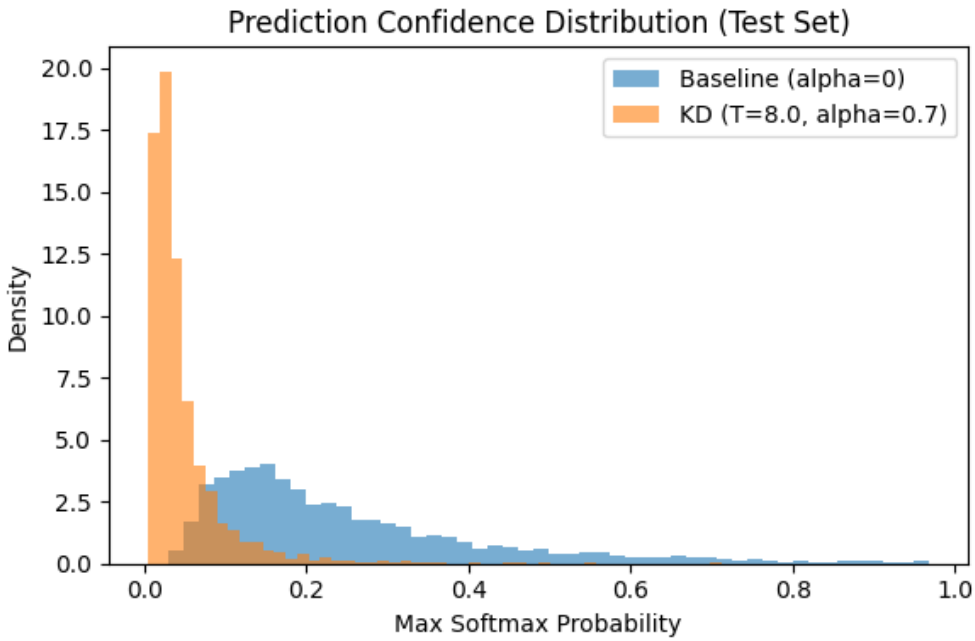
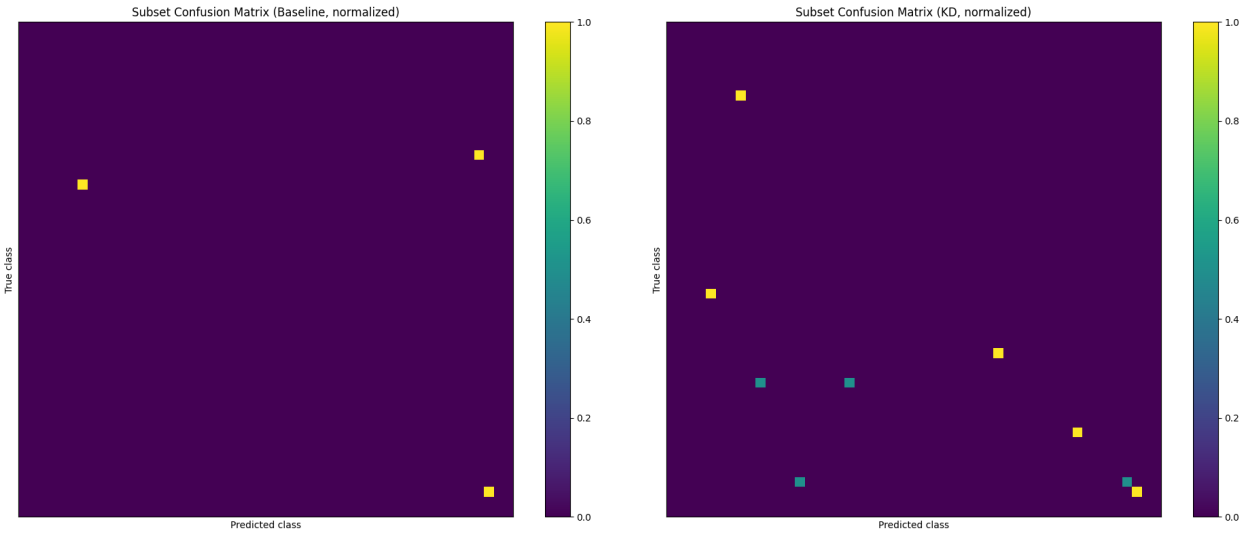


Figure 10: Distribution of maximum softmax confidence scores on the test set. The KD model produces better-calibrated and less over-confident predictions compared to the EEG-only baseline.



(a) Baseline (row-normalized)

(b) KD (row-normalized)

Figure 11: Row-normalized confusion matrices on a representative subset of 50 classes. Compared to the baseline, the KD-trained model shows clearer diagonal structure and reduced systematic confusions.

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